

StockSense — Complete Product & Technical Documentation

Live Document — Updated automatically as the product evolves.

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1. Product Overview

StockSense is an AI-powered stock prediction and portfolio intelligence platform built for Indian and US equity markets. It combines institutional-grade quantitative signals with a consumer-friendly interface to deliver actionable BUY / HOLD / SELL signals with full explainability.

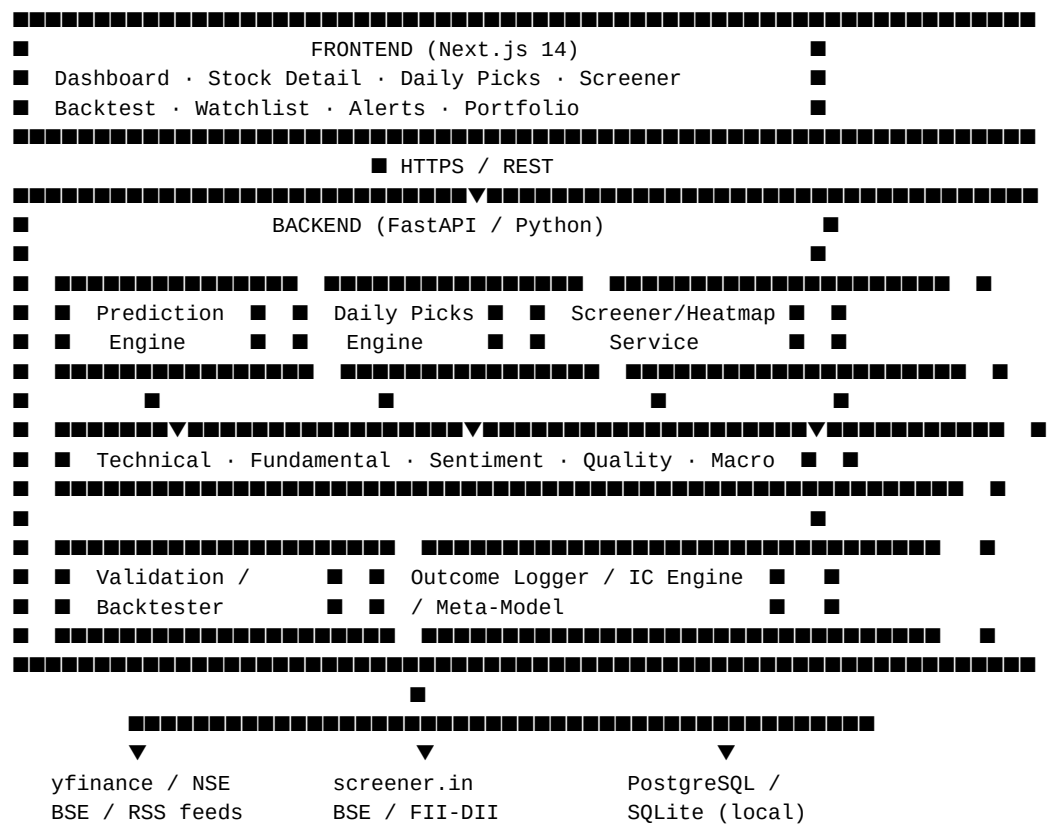
What StockSense Does

- Generates **BUY / HOLD / SELL signals** for Nifty 100, US large-cap, and top cryptocurrencies
- Delivers **Daily Picks** every morning at 9 AM IST — top 5 BUY ideas per horizon (short / medium / long)
- Shows **why** every signal was generated — factor breakdown, confidence scores, reasoning bullets
- Provides **trade levels** — entry zone, stop-loss, and target price with R:R ratio
- Runs a **learning engine** — tracks prediction outcomes and retrains factor weights weekly
- Supports **screener, backtest, watchlist, and alerts**

Supported Markets

Market	Universe	Horizons
India (NSE)	Nifty 100	Short (1–5 days), Medium (2–4 weeks), Long (3–6 months)
US	S&P 500 large-caps	Short, Medium, Long
Crypto	BTC, ETH, BNB, SOL, XRP, DOGE, ADA, AVAX, LINK, DOT	Short, Medium, Long

2. Architecture Overview



Tech Stack

Layer	Technology
Frontend	Next.js 14, React, TypeScript, TailwindCSS, Recharts
Backend	Python 3.11, FastAPI, Uvicorn
Auth	Supabase (JWT)
Database	PostgreSQL (production), SQLite (local)
Cache	In-memory (TTL-based), disk (picks_cache.json)
Hosting	Render.com (backend), Vercel (frontend)
Automation	GitHub Actions (cron jobs)

3. Data Sources

Source	Data Provided	Frequency	Used For
yfinance	Price, OHLCV, P/E, ROE, FCF, beta, analyst targets	Real-time	All markets
screener.in	10-year financials, ROCE, CAGR, promoter %, pledge %	Daily	India fundamentals
BSE API	Fundamentals for renamed / merged stocks	Daily	India fallback
NSE FII/DII API	Daily institutional flows (■ Cr)	Daily	India quality signal
NSE Pledge API	Promoter pledge % (quarterly disclosure)	Quarterly	India risk signal
Yahoo Finance RSS	News headlines per symbol	Real-time	Sentiment
Google News RSS	{symbol} stock India search results	Real-time	Sentiment fallback
Economic Times RSS	India economy & market news	Real-time	India sentiment
MoneyControl RSS	Stock & sector news	Real-time	India sentiment
yfinance macro	S&P 500, VIX, Crude, Gold, USD/INR, Nifty IT/Bank	15-min cache	Global macro

Data Fallback Chain (India)

yfinance → screener.in (if <5 key fields) → BSE API (if still sparse)

4. Core Prediction Engine

File: [backend/services/prediction_engine.py](#)

Signal Generation Formula

Composite Score = (Tech × W_tech) + (Fund × W_fund) + (Sent × W_sent)
+ Global Macro Adjustment
+ Analyst Consensus Adjustment
+ 52-Week Position Adjustment
+ Quality Factor Adjustment
+ Rounding Adjustment
- Risk Penalty

All component scores are on a **0–100 scale** (50 = neutral). The composite is also 0–100.

Signal Thresholds

Composite Score	Signal	Score Band	Confidence Calculation
≥ 90	BUY	Exceptional Opportunity	$(\text{score} - 60) / 40 \times 100\%$
≥ 75	BUY	Strong Buy Candidate	$(\text{score} - 60) / 40 \times 100\%$
≥ 60	BUY	Good Watchlist Stock	$(\text{score} - 60) / 40 \times 100\%$
45 – 59	HOLD	Neutral — Monitor	$50 - \text{abs}(\text{score} - 52) \times 2$
< 45	SELL	Avoid	$(45 - \text{score}) / 45 \times 100\%$

Dynamic Weights (Horizon × Volatility × Regime)

Base weights by horizon:

Horizon	Technical	Fundamental	Sentiment
Short (1–5 days)	70%	15%	15%
Medium (2–4 weeks)	40%	45%	15%
Long (3–6 months)	15%	75%	10%

Volatility modulation (applied on top of base weights):

Volatility Level	Annualised Vol	Technical	Fundamental
High	> 35%	Reduced to 10%	Boosted
Normal	15–35%	Base	Base
Low	< 15%	Boosted to 75%	Reduced

Regime modulation:

- **BULL regime:** Boost technical, reduce fundamental

- **BEAR regime:** Boost fundamental, reduce technical
- **SIDEWAYS:** No modulation (enables mean-reversion trades)

Prediction Caching

- **TTL:** 15 minutes per `(symbol:market:horizon)` key
- **Max size:** 300 entries (Render 512 MB free-tier limit)
- **Eviction:** LRU — oldest entry dropped when full
- **Async pattern:** First request returns HTTP 202 (computing); client polls every 5s; result cached on completion

5. Technical Analysis Module

File: `backend/services/technical_indicators.py`

Indicators Computed

Category	Indicator	Parameters
Momentum	RSI	14-period
	Stochastic RSI	14-period
	Williams %R	14-period
	CCI	20-period
	Stochastic Oscillator	K=14, D=3
Trend	MACD	12/26/9 EMA
	EMA	20, 50, 200 periods
	ADX + DI+/DI−	14-period
Volatility	Bollinger Bands	20-period, 2σ
	ATR	14-period
Volume	OBV	Rolling
	Volume SMA	20-period
	VWAP	20-day rolling

Scoring Rules

RSI:

- < 30 (oversold): **+15 pts**
- 30–45 (recovering): **+7 pts**
- 60–70 (elevated): **−7 pts**
- > 70 (overbought): **−15 pts**

MACD:

- Above signal line: **+12 pts**
- Below signal line: **−12 pts**

EMA Trend:

- Price > EMA200 (bull market structure): **+10 pts**
- EMA20 > EMA50 (golden cross zone): **+8 pts**
- EMA20 < EMA50 (death cross zone): **−8 pts**

ADX (Trend Strength):

- ADX > 25 AND +DI > −DI (strong uptrend): **+10 pts**
- ADX > 25 AND −DI > +DI (strong downtrend): **−10 pts**
- ADX ≤ 25 (sideways): **0 pts**

Bollinger Bands (%B):

- Near lower band (< 0.1): **+8 pts** (oversold)
- Near upper band (> 0.9): **−8 pts** (overbought)

Volume Confirmation:

- Up-day volume > down-day volume: **+8 pts**
- Down-day volume > up-day volume: **−8 pts**

Candlestick Patterns:

Pattern	Points	Signal
Hammer	+5	BUY
Morning Star	+5	BUY
Bullish Engulfing	+5	BUY
Shooting Star	−5	SELL
Evening Star	−5	SELL
Bearish Engulfing	−5	SELL

Technical Signal Output

- Score ≥ 58: **BUY**
- Score ≤ 42: **SELL**
- 42–58: **HOLD**

6. Fundamental Scoring Module

File: `backend/services/prediction_engine.py` → `_fundamental_score()`

Quality Gate (Hard Rejection)

A stock is flagged **REJECTED** before scoring if **any** of these apply:

- ROE < −10% (severely destroying shareholder value)
- Profit Margin < −15% (deeply loss-making)

- Non-positive Operating Cash Flow on medium/long horizons (core business not generating cash)
- D/E ratio > 500% (extreme leverage, non-financial sector only)

Previously ROE AND margin both had to be negative simultaneously — this was too lenient and has been corrected.

Scoring Architecture (Per-Category Budgets)

The fundamental score uses **six independent capped buckets**, each scored separately, then summed with a base of 50. This prevents any single dimension from dominating and ensures the final score has meaningful discrimination across the full 0–100 range.

Bucket	Cap	What It Measures
Valuation	±15	P/E, P/B ratios vs market-calibrated thresholds
Profitability	±15	ROE, ROCE, profit margins
Growth	±15	Revenue + earnings CAGR (counted once via longest available window)
Balance Sheet	±10	D/E, OCF quality, Altman Z-Score, Sloan accruals
Governance	±10	Promoter holding, FII/DII flows, promoter pledge
Banking	±10	Net NPA, NIM (fires only for banks/NBFCs)

Total possible range: $50 \pm 65 \rightarrow$ clamped to $[0, 100]$

Valuation bucket (cap ±15)

Metric	Threshold (India / US)	Points
P/E	< 18 IN / < 15 US (cheap)	+8
P/E	< 30 IN / < 25 US (fair)	+3
P/E	> 55 IN / > 50 US (expensive)	−8
P/B	< 2.5 IN / < 2.0 US	+4
P/B	> 8.0 IN / > 6.0 US	−4

Profitability bucket (cap ±15)

Metric	Threshold	Points
ROE	> 20%	+7
ROE	10–20%	+3
ROE	< 0%	−7
Profit Margin	> 20%	+5

Metric	Threshold	Points
Profit Margin	< 0%	−5
ROCE	> 20%	+6
ROCE	12–20%	+2
ROCE	< 6%	−4

Growth bucket (cap ±15) — revenue and earnings each counted once

Revenue uses 3Y CAGR (screener.in) if available, else TTM YoY (yfinance):

Metric	Threshold	Points
3Y Revenue CAGR	> 15%	+7
3Y Revenue CAGR	8–15%	+3
3Y Revenue CAGR	< 0%	−5
TTM Revenue Growth (fallback)	> 20%	+7
TTM Revenue Growth	5–20%	+3
TTM Revenue Growth	< −5%	−5

Earnings uses longest available: 5Y CAGR (long horizon) → 3Y CAGR → TTM EPS growth:

Metric	Threshold	Points
5Y Profit CAGR (long only)	> 18%	+6
3Y Profit CAGR	> 20%	+6
3Y Profit CAGR	> 10%	+3
3Y Profit CAGR	< −10%	−5
TTM EPS Growth (fallback)	> 20%	+5
TTM EPS Growth	< −10%	−5
Quarterly PAT trend	Accelerating	+3
Quarterly PAT trend	Decelerating	−3

Balance Sheet bucket (cap ±10)

Metric	Threshold	Points
D/E	> 300%	−7
D/E	150–300%	−3
D/E	< 50%	+3
Operating CF (screener)	Negative	−5
Operating CF 3Y growth	> 30%	+4

Metric	Threshold	Points
Operating CF 3Y growth	Positive	+2
Altman Z-Score	Safe zone (medium/long)	+3
Altman Z-Score	Grey zone	−4
Altman Z-Score	Distress zone	−8
Sloan Accruals ratio	< −5% (cash-backed)	+3
Sloan Accruals ratio	> 10% (manipulation risk)	−5

Governance bucket (cap ±10) — India only

Metric	Threshold	Points
FII + DII combined	> 50%	+4
FII + DII combined	25–50%	+2
DII quarterly trend	Up > 3% (MF accumulation)	+3
DII quarterly trend	Down > 3%	−3
FII quarterly trend	Up > 3%	+2
FII quarterly trend	Down > 3%	−2
Promoter holding	> 55%	+2
Promoter holding	< 25%	−2
Promoter trend	Up > 2% (insider buying)	+3
Promoter trend	Down > 3% (insider selling)	−4
Promoter pledge	> 50%	−8
Promoter pledge	25–50%	−5
Promoter pledge	10–25%	−2
Promoter pledge	0%	+2

Banking bucket (cap ±10) — fires only for banks/NBFCs

Metric	Threshold	Points
Net NPA	> 3%	−7
Net NPA	1.5–3%	−3
Net NPA	< 0.5%	+4
NIM	> 4%	+4
NIM	< 2%	−3

7. Sentiment Analysis Module

File: `backend/services/news_sentiment.py`

Two-Layer Scoring

Layer 1 — Financial Lexicon (60% weight):

- ~80 domain-specific phrases with pre-calibrated scores
- Examples: "beat expectations" → +0.75, "profit warning" → -0.80, "upgrade" → +0.65
- Designed to override generic sentiment on financial language

Layer 2 — VADER Sentiment (40% weight):

- Title sentiment (70%) + Description sentiment (30%)
- Standard NLP library tuned for social media / news

Final blend: $\text{score} = 0.60 \times \text{lexicon} + 0.40 \times \text{VADER}$

News Sources (per prediction)

Source	Articles	Priority
Yahoo Finance RSS (<code>{symbol}</code> , <code>{symbol}.NS</code>)	10	Primary
Google News RSS (<code>{symbol} stock India</code>)	10	Secondary
Economic Times RSS	10	India supplement
MoneyControl RSS	10	India supplement

Cache TTL: 10 minutes per symbol

Sentiment Classification

Score	Label
$\geq +0.05$	BULLISH
-0.05 to $+0.05$	NEUTRAL
≤ -0.05	BEARISH

Score conversion: $\text{sentiment_score} = 50 + (\text{bullish} - \text{bearish}) / (\text{bullish} + \text{bearish}) \times 50$

Neutral articles are excluded from the denominator. Previously neutrals diluted the score — 5 bullish + 5 neutral incorrectly scored the same as 5 bullish + 5 bearish. Now only labelled articles (bullish + bearish) count.

When no news available: Returns neutral (50), redistributes weight to technical + fundamental, sets `data_available = False` flag.

8. Global Macro Context Module

Macro Indicators Tracked

Indicator	Ticker	What It Signals
S&P 500	^GSPC	US demand → IT/pharma export tailwind
NASDAQ	^IXIC	Tech hiring, demand for Indian IT services
VIX	^VIX	FII flows: high VIX → FII outflows from India
USD/INR	INR=X	Weak INR → pharma/IT revenue boost; import cost rise
Crude Brent	BZ=F	Oil import cost, OMC margins
Gold	GC=F	Jewelry demand (Titan, Muthoot)
Nifty IT Index	^CNXIT	Sector momentum for IT stocks
Nifty Bank Index	^NSEBANK	Banking sector rotation signal

Cache TTL: 15 minutes (fetched in parallel)

Stock-Specific Macro Sensitivity Map

Over 100 Nifty stocks mapped to their macro sensitivities:

Stock Category	Tailwind Factors	Headwind Factors
IT (TCS, INFY, WIPRO, HCL)	USD/INR weakness, S&P 500 up, NASDAQ up	INR strengthening
Pharma (SUNPHARMA, DRREDDY)	USD/INR weakness, S&P 500 up	—
OMCs (BPCL, IOC)	Crude down	Crude up (margin squeeze)
Oil producers (ONGC)	Crude up	Crude down
Paints (ASIANPAINT, PIDILITIND)	—	Crude up (TiO2 input cost)
Banks (HDFCBANK, ICICIBANK)	—	VIX up (FII sensitivity)
Jewelry (TITAN)	Gold stable	Gold up (input cost)

Stock Adjustment Calculation

stock_adj = $\sum \text{sensitivity}_i \times (\text{factor}_i - \text{benchmark}_i)$
Tailwind factors: +2 to +4 points
Headwind factors: -2 to -4 points

9. Quality Factors Module

File: [backend/services/quality_factors.py](#)

Applied for **India (Nifty 100)** only, on medium and long horizons.

10 Dimensions of Quality

1. Earnings Revisions (Weight: 12–14%)

- EPS surprise trend (beat vs miss last 4 quarters): ± 16 pts
- Analyst upgrade/downgrade momentum: ± 8 pts
- Forward PE compression vs trailing PE: ± 8 pts

2. Institutional Ownership (Weight: 5–6%)

- Holdings > 50%: +14 pts
- Holdings 30–50%: +8 pts
- Holdings 15–30%: +3 pts
- Holdings < 5%: –5 pts
- Institution count > 300: +6 pts; < 20: –4 pts

3. Institutional Flow Proxy / MF Trend (Weight: 5–13%)

Blends price-volume signals (OBV trend, MFI, accumulation pattern) with real NSE FII/DII data:

- 60% price-volume proxy + 40% real FII/DII flows (when available)

4. Relative Strength (Weight: 7–15%)

Stock return vs NIFTY 50 over 1M, 3M, 6M:

- Outperform > 10%: +12 pts
- Outperform 4–10%: +6 pts
- Underperform > 10%: –12 pts
- Underperform 4–10%: –6 pts

5. Sector Strength (Weight: 7–15%)

Sector index momentum vs NIFTY 50:

- Sector outperform > 5%: +16 pts
- Sector outperform 2–5%: +8 pts
- Sector underperform > 5%: –14 pts
- Sector underperform 2–5%: –7 pts

6. Valuation Quality (Weight: 8–17%)

Multi-dimensional valuation scoring:

- PEG < 0.75: +16 pts; > 2.5: –12 pts
- EV/EBITDA vs sector: ± 10 pts at major discount/premium
- Sector-relative PE: ± 12 pts at 30% discount
- P/B for banks < 1.0: +12 pts; > 4.0: –8 pts
- Analyst target upside > 30%: +10 pts (margin of safety)

7. Risk Management (Weight: 10%)

- Max Drawdown 12M < –10%: +14 pts; < –40%: –14 pts

- Volatility Percentile bottom 25%: +10 pts; top 20%: –10 pts
- Sharpe Ratio > 1.5: +12 pts; < 0: –10 pts
- Downside Deviation < 10%: +6 pts; > 30%: –6 pts

8. Corporate Actions (Weight: 3–10%)

- Dividend payer 5Y+: +8 pts
- Growing dividends: +6 pts
- Payout ratio 0–40%: +4 pts; > 80%: –6 pts
- Active buyback: +8 pts
- Share dilution: –5 pts

9. Liquidity (Weight: 4–8%)

- Market cap ■2T+: +10 pts; < ■10B: –8 pts
- Avg daily volume > 5M: +8 pts; < 100K: –8 pts
- Beta 0.5–1.2: +4 pts; > 2.0: –5 pts

10. Quality Metrics (Weight: 4–12%)

Piotroski F-Score (9-point scale):

Point	Criterion
P1	ROA > 0
P2	Operating Cash Flow > 0
P3	ROA improving YoY
P4	Accruals: cash earnings > accrual earnings
P5	Leverage decreasing
P6	Current ratio improving
P7	No share dilution
P8	Gross margin expanding
P9	Asset turnover improving

- Score ≥ 8: +20 pts
- Score 6–7: +10 pts
- Score 4–5: 0 pts
- Score < 4: –12 pts

ROIC (Return on Invested Capital):

- > 20%: +14 pts
- 12–20%: +8 pts
- 6–12%: +2 pts
- < 6%: –8 pts

Horizon-Based Weighting

Dimension	Short	Medium	Long
Earnings Revisions	13%	14%	12%
Institutional Ownership	5%	6%	6%
MF/FII Flow	6%	8%	10%
Relative Strength	15%	11%	7%
Sector Strength	15%	11%	7%
Valuation Quality	8%	13%	17%
Risk Management	10%	10%	10%
Corporate Actions	3%	6%	10%
Liquidity	8%	6%	4%
Quality Metrics	4%	6%	12%
Flow Proxy	13%	9%	5%

10. Risk Penalty Framework

Applied **after** all signal scoring. Never adds points — only subtracts (risk override).

Risk Factor	Condition	Penalty
High leverage	D/E > 300%	−8 pts
Elevated leverage	D/E 200–300%	−4 pts
High beta	Beta > 2.0	−6 pts
Elevated beta	Beta 1.6–2.0	−3 pts
Negative FCF	FCF < 0	−5 pts
Negative ROE	ROE < 0	−5 pts
Poor risk profile	Risk score < 35	−5 pts
Earnings volatility	CV > 0.5	−4 pts

Maximum penalty capped at −30 pts (prevents extreme scores on genuinely bad risk/reward stocks).

11. Confidence Engine

Answers: "How much should you trust this signal?"

Five Components

Component	What It Measures	Weight (Full)	Weight (Bootstrap)
Data Completeness	% of key fundamental fields present	25%	31.25%
Factor Agreement	% of factors agreeing with signal direction	25%	31.25%
Earnings Stability	Quality earnings_revision sub-score	15%	18.75%
Regime Certainty	BULL/BEAR vs SIDEWAYS trend strength	15%	18.75%
Historical Factor Reliability	Live IC values (needs 60+ outcomes)	20%	0% → fallback 50

Sector-Aware Field Sets

Indian Non-Financial: PE, ROE, revenue growth, D/E, margin, beta (yfinance) + 3Y CAGR, ROCE, FII, promoter (screener) = 10 fields

Indian Financial (banks/NBFCs): PE, ROE, revenue growth, profit margin, EPS growth, beta (yfinance) + 3Y CAGR, FII, promoter (screener) = 10 fields

US Stocks: PE, ROE, revenue growth, D/E, margin, EPS growth, FCF, beta, ROCE = 13 fields

Confidence Bands

Score	Band
≥ 80	HIGH
60–79	MEDIUM
< 60	LOW

Percentile Context

Score	Label
≥ 80	Top 10% of all Nifty predictions
≥ 72	Top 20%
≥ 65	Top 35%
≥ 58	Top 50%
≥ 50	Below average
< 50	Low range

12. Target Price & Trade Levels

File: `backend/services/prediction_engine.py` → `_target_price()`, `_trade_levels()`

Target Price

Horizon	Method
Short	ATR × 2.5 × confidence factor — moves 1–5 day magnitude
Medium	Analyst target (70%) + price projection (30%); BUY floor: <code>max(blend, price×1.05)</code>
Long	<code>analyst_target × (1+EPS_growth)²</code> ; BUY floor: <code>max(target, price×1.15)</code>

Trade Levels

Take Profit = Model Target Price (consistent with signal)
Stop Loss = ATR-based, adjusted to maintain minimum R:R of 1.5×
Entry Zone = [price − 0.3×ATR, price + 0.1×ATR] for BUY
 [price − 0.1×ATR, price + 0.3×ATR] for SELL
R:R Ratio = (target − price) / (price − stop_loss)

ATR multipliers by horizon:

Horizon	Base SL	Floor SL	Max SL
Short	1.5× ATR	1× ATR	25% of price
Medium	3× ATR	2× ATR	25% of price
Long	5× ATR	3× ATR	25% of price

13. Daily Picks Engine

File: `backend/services/daily_picks.py`

Execution Schedule

- Triggered: **Every weekday at 9:00 AM IST** (via GitHub Actions → POST `/api/picks/generate`)
- Generation time: ~10 minutes
- Results cached to disk: `backend/picks_cache.json`
- API response: Instant (reads from cache)

9-Phase Pipeline

Phase 0 — Outcome Resolution

- Compare previous predictions against actual forward returns (1-day / 63-day / 252-day)
- Update outcome logger database with direction hits (correct/incorrect)
- Feed data into IC engine for retraining

Phase 1 — Universe Screening

- Run full prediction engine on all Nifty 100 stocks
- Parallelised: 2 workers via `ThreadPoolExecutor`
- Returns raw factor scores for all stocks (enables cross-sectional z-scoring)

Phase 2 — Regime Detection

- KMeans clustering (4 clusters) on global macro features: VIX, S&P 500 return, crude, gold, USD/INR
- Classifies market into: **BULL_CALM**, **BULL_VOLATILE**, **BEAR_CALM**, **BEAR_PANIC**
- Returns regime label + weight multipliers for IC adjustment

Phase 3 — IC Weight Computation

- If < 60 outcome pairs: use academic prior weights
- Short: tech=0.055, fund=0.018, sentiment=0.042, quality=0.032
- If ≥ 60 pairs: Bayesian shrinkage blend of live IC + prior:
- $\text{weight} = \text{live_weight} \times \text{live_ic} + (1 - \text{live_weight}) \times \text{prior}$
- Apply regime multipliers: BULL boosts tech/sentiment; BEAR boosts fundamental/quality

Phase 4 — Z-Score Normalisation & Alpha

- Cross-sectional z-scoring: $z_i = (\text{score}_i - \text{mean}) / \text{std}$
- Combined alpha: $\sum \text{ic_weight}_k \times z_k$
- Meta-model alpha (requires 180+ outcomes across 60 stocks × 3 horizons):
- Inputs: tech_z, fund_z, sentiment_z, quality_z, combined_alpha, regime_id
- Output: predicted return %

Phase 5 — Pick Selection

- Rank by alpha score (meta_alpha if available, else combined_alpha)
- Select top 5 **BUY** signals per horizon (composite score ≥ 60)
- Minimum 1 pick per horizon
- Empty picks from a prior run (0 BUY signals) are NOT treated as "complete" — startup catch-up will retry on next deploy

Phase 6 — Portfolio Optimisation

- Fetch 6-month daily returns for selected picks
- Covariance estimation: Ledoit-Wolf shrinkage at 25%
- Optimise: $\max (\alpha \times w - \lambda \times w^T \Sigma w)$ via SLSQP
- Constraints: $\Sigma w = 1.0, 0 \leq w_i \leq 0.40$ (max 40% per position)
- Risk aversion λ : doubled in BEAR_PANIC regime
- Fallback (if scipy unavailable): iterative alpha-proportional weights that correctly enforce the 40% cap

Phase 7 — Logging

- Log to PostgreSQL (if `USE_POSTGRES=1`) or SQLite
- Stored fields: factor z-scores, combined_alpha, meta_alpha, signal, price, horizon, regime

Phase 8 — Weight Adaptation (background)

- Retrain IC engine with new outcomes
- Update meta-model if sufficient data
- Recalibrate regime clustering

- Runs in daemon thread (non-blocking)

14. Backtesting & Validation Engine

File: `backend/services/validation_engine.py`, `backend/services/backtester.py`

Walk-Forward Methodology

For each business day t in Nifty 100 history:

- Fetch price data available **only before** time t
- Compute prediction at t using that data
- Measure actual forward return at $t + h$ ($h = 7 / 63 / 252$ days)
- Compare predicted signal vs actual direction

Known limitation (next session): Technical indicators (EMA, MACD, OBV) are currently computed on the full historical DataFrame before slicing per date — a form of look-ahead bias in the backtester. This inflates reported validation hit rates. A full fix (recompute indicators per window) is planned for Session 5.

Metrics Computed

Metric	Definition
Hit Rate	% of BUY/SELL calls that were directionally correct
Avg Return on BUY	Mean forward return when signal was BUY
Sharpe Ratio	$(\text{avg_return} - \text{risk_free}) / \text{std_return}$ on BUY calls
vs Benchmark	Alpha over NIFTY 50 buy-and-hold
Score Calibration	Hit rate by score bucket (60–70, 70–80, 80–100)

Execution

- **Weekly:** Sunday 7:30 AM IST (via GitHub Actions)
- **Duration:** ~40 minutes (medium + long horizons)
- **Storage:** PostgreSQL `val_runs` + `val_signals` tables
- **Retention:** 365 days

Forward Return Windows

Horizon	Forward Return Used	Outcome Resolution Wait
Short	return_5d (5 trading days)	3 calendar days
Medium	return_20d (20 trading days $\approx$ 1 month)	30 calendar days
Long	return_60d (60 trading days $\approx$ 3 months)	90 calendar days

Partial returns are never logged — the outcome logger returns None if fewer than the required trading days have elapsed. This prevents truncated returns from contaminating IC training data.

Validation BUY Threshold

Validation uses the same threshold as the live prediction engine: **composite** ≥ 60 = **BUY** for all horizons. Previously the thresholds were mismatched (validation used 65, live used 60), making validation metrics unmeasurable against the actual model.

Indicative Score Calibration (Nifty 100, Medium-Term)

Score Bucket	Hit Rate	Beat Benchmark	Avg Alpha
80–100 (Exceptional/Strong BUY)	~68%	~62%	+4.2%
75–79 (Strong BUY)	~62%	~55%	+2.8%
60–74 (Good Watchlist BUY)	~58%	~50%	+1.5%
45–59 (HOLD)	~52%	~48%	+0.3%

Note: These figures reflect the post-Session 4 thresholds. Historical validation data accumulated under the old 70-threshold is being re-calibrated.

15. Crypto Prediction Module

File: `backend/services/crypto_engine.py`

Supported Assets

BTC, ETH, BNB, SOL, XRP, DOGE, ADA, AVAX, LINK, DOT

Signals Used (No Fundamentals)

Signal	Weight
Technical indicators (full suite)	Primary
Fear & Greed proxy (30-day vol / 90-day vol)	Secondary
On-chain proxy (price-volume accumulation)	Secondary
News sentiment (Google News + CoinTelegraph)	Supporting
Macro sensitivity (BTC/ETH correlated to S&P, VIX)	Adjustment

Quality factors and dynamic fundamental weights are not applied — crypto lacks P/E, ROE, cash flow data. Momentum and sentiment dominate.

16. Screener & Universe Management

File: [backend/services/screener_service.py](#), [screener_data.py](#)

Screener Filters

Filter	Type	Description
Market	US / IN	Filter by market
Signal	BUY / HOLD / SELL	Filter by signal
Min Market Cap	■ Cr	Size filter
Max P/E	Ratio	Valuation filter
Min ROE	%	Profitability filter
Sector	IT / Banking / Pharma / etc.	Sector filter

Heatmap

- Groups stocks by sector; sorted by sector avg change% (best sectors on top)
- **India: 25 sectors** — Banking, IT, Auto, Pharma, Energy, FMCG, Finance, Healthcare, Insurance, Chemicals, Cement, Metal & Mining, Defence, Realty, Telecom, Consumer Disc, Hotels & Travel, Food & Beverage, Media & Entmt, Textiles, Agro & Chemicals, Logistics, Paints, Infra, Capital Goods, Power, EV & New Energy
- **US: 29 sectors** — Mega Cap Tech, Semiconductors, Cloud & SaaS, Cybersecurity, Fintech, Finance, Insurance, Healthcare, Biotech, Med Devices, Energy, Clean Energy, EV, Consumer Disc, Consumer Stap, E-commerce, Social Media, Streaming & Media, Gaming, Aerospace & Defence, Industrials, Airlines, Cruise & Hotels, Restaurants, Retail, Telecom, Utilities, Realty, Materials, Crypto & Blockchain
- Up to **15 stocks per sector** (MAX_STOCKS = 15)
- Primary data source for India sectors: **NSE sector index APIs** ([SECTOR_TO_NSE_INDEX](#) mapping); Yahoo Finance used as fallback for symbols not in NSE indices and as primary source for all US sectors
- Grey tiles = symbol has no live data; all 353 India symbols audited and bad tickers corrected (Session 5)
- Colour-coded by performance (green/red intensity); loading status badge (Fetching / Refreshing / Live)

Top Movers

- Fetched in real-time from yfinance
 - Ranked by absolute % change
 - Separate for US, India, Crypto
-

17. Paper Trading Module

File: [backend/api/routers/paper_trading.py](#), [frontend/src/app/paper-trading/page.tsx](#)

A simulated trading environment where users can test stock calls without real money. All trades persist in PostgreSQL and survive Railway restarts.

Database Schema

```
paper_portfolio (user_id TEXT PK, session_id TEXT, cash NUMERIC, updated_at TIMESTAMPTZ)
paper_trades    (id SERIAL PK, user_id TEXT, session_id TEXT, symbol TEXT, market TEXT,
```

```
quantity INT, entry_price NUMERIC, exit_price NUMERIC,  
stop_loss NUMERIC, target_price NUMERIC, status TEXT,  
signal TEXT, horizon TEXT, opened_at TIMESTAMPTZ, closed_at TIMESTAMPTZ)
```

User Model

- Trades are scoped to **Supabase** `user_id` (stable UUID from `useAuth().user.id`)
- No more `session_id` / `localStorage` dependency — trades persist across all browsers and devices
- Starting virtual cash: **■1,00,000 / \$10,000** (depending on market)
- All positions are long only (no shorting)

Trade Lifecycle

Open Trade → `entry_price` captured at trade time, keyed to `user_id`
Live Price → fetched real-time via `yfinance` / `NSE`
Unrealised P&L = $(\text{live_price} - \text{entry_price}) \times \text{quantity}$
Close Trade → `exit_price` set, `status` = 'CLOSED'
Realised P&L = $(\text{exit_price} - \text{entry_price}) \times \text{quantity}$

Stop Loss & Target Price

- Optionally set per trade via inline edit (↵ icon)
- ATR-based defaults pre-filled when placing a trade from the stock detail page
- Both values always shown per position with % from entry
- UI highlights rows where price is within 2% of stop (yellow) or target (green)
- The Buy button is **not blocked** by AI prediction loading — trade is placeable immediately; stop loss/target suggestions update in the background

API Endpoints

Endpoint	Method	Description
<code>/api/paper-trading/portfolio</code>	GET	Portfolio summary (pass <code>user_id</code> as query param)
<code>/api/paper-trading/buy</code>	POST	Open a new position (body: <code>user_id</code> , symbol, market, qty, price, ...)
<code>/api/paper-trading/sell/{trade_id}</code>	POST	Close a position (body: <code>user_id</code> , price)
<code>/api/paper-trading/trade/{trade_id}</code>	PATCH	Edit <code>stop_loss</code> / <code>target_price</code> (body: <code>user_id</code> , ...)
<code>/api/paper-trading/reset</code>	POST	Reset portfolio to starting cash (query: <code>user_id</code>)

18. Alerts System

File: `backend/api/routers/alerts.py`, `frontend/src/app/alerts/page.tsx`

Price-level alerts that trigger when a stock crosses a target price. All alerts persist in PostgreSQL.

Database Schema

```
price_alerts (id TEXT PK, user_id TEXT, symbol TEXT, market TEXT,  
              target_price NUMERIC, direction TEXT CHECK('above','below'),  
              triggered BOOL DEFAULT FALSE, created_at TIMESTAMPTZ, triggered_at TIMESTAMPTZ)
```

Alert Check Logic

- Frontend polls live quote every 60 seconds
- Compares `live_price` vs `target_price` by direction (`above` / `below`)
- On trigger: PATCH alert to `triggered = true`, stores `triggered_at`
- Triggered alerts shown with timestamp; can be reset or deleted

API Endpoints

Endpoint	Method	Description
<code>/api/alerts/{user_id}</code>	GET	All alerts for user
<code>/api/alerts/{user_id}</code>	POST	Create new alert
<code>/api/alerts/{user_id}/{alert_id}</code>	PATCH	Update (reset triggered, edit price)
<code>/api/alerts/{user_id}/{alert_id}</code>	DELETE	Remove alert

19. API Reference

Prediction

Endpoint	Method	Description
<code>/api/predictions/{symbol}</code>	GET	Get prediction (async, 202 while computing)
<code>/api/predictions/debug/state</code>	GET	Internal cache/thread state (debug only)

Query params: `market` (US/IN/CRYPTO), `horizon` (short/medium/long)

Daily Picks

Endpoint	Method	Description
<code>/api/picks/daily</code>	GET	Today's cached picks (instant)
<code>/api/picks/generate</code>	POST	Trigger pick generation (secret-protected)

Endpoint	Method	Description
/api/picks/performance	GET	Live P&L of past picks vs benchmark

Query params for performance: `horizon` (short/medium/long), `window_days` (default 90)

Screener

Endpoint	Method	Description
/api/screener/filter	GET	Filter universe by fundamentals
/api/screener/heatmap	GET	Sector heatmap
/api/screener/top-movers	GET	Top 10 movers by % change
/api/screener/crypto-movers	GET	Top 10 crypto movers

Backtest & Validation

Endpoint	Method	Description
/api/backtest/run	POST	Single-stock walk-forward backtest
/api/validation/run	POST	Full universe validation (background)
/api/validation/status	GET	Validation progress
/api/validation/results	GET	Validation metrics by horizon

Watchlist

Endpoint	Method	Description
/api/watchlist/{user_id}	GET	All saved stocks
/api/watchlist/{user_id}	POST	Add stock
/api/watchlist/{user_id}/{symbol}	DELETE	Remove stock

Alerts

Endpoint	Method	Description
/api/alerts/{user_id}	GET	All price alerts
/api/alerts/{user_id}	POST	Create alert
/api/alerts/{user_id}/{alert_id}	PATCH	Update alert (reset / edit)
/api/alerts/{user_id}/{alert_id}	DELETE	Delete alert

Paper Trading

Endpoint	Method	Description
/api/paper-trading/portfolio	GET	Portfolio summary (?user_id=)
/api/paper-trading/buy	POST	Open position
/api/paper-trading/sell/{trade_id}	POST	Close position
/api/paper-trading/trade/{trade_id}	PATCH	Edit SL / target
/api/paper-trading/reset	POST	Reset portfolio (?user_id=)

Infrastructure

Endpoint	Method	Description
/health	GET	Health check (keep-alive ping)

20. Frontend Pages & Components

Pages

Page	Route	Description
Landing	/ (unauthenticated)	Public marketing page — features, how-it-works, CTA adapts to login state
Dashboard	/dashboard	Movers (US/IN/Crypto), market status, live index bar with loading badge
Stock Detail	/stock/:symbol	Prediction, trade levels, factor breakdown, news, chart
Daily Picks	/picks	Top BUY ideas by horizon, portfolio weights, trust layer
Screener	/screener	Filter and explore the universe
Backtest	/backtest	Single-stock historical walk-forward test
Heatmap	/heatmap	Sector colour-coded snapshot (IN / US) with loading badge
Watchlist	/watchlist	Saved stocks with live prices and change% — user-scoped
Alerts	/alerts	Price alerts with live trigger detection — user-scoped

Page	Route	Description
Portfolio	/portfolio	Holdings with BUY/HOLD/SELL per position
Paper Trade	/paper-trading	Simulated trading — open/close positions, P&L — user-scoped
Validation	/validation	Hit rate, Sharpe, alpha vs benchmark

Key Components

Component	Description
FactorAttributionWaterfall	Score decomposition bar chart; click to drill down
ConfidenceMeter	Colour-coded progress bar (0–100)
ConfidenceBreakdown	SVG gauge + 5 confidence bars with tooltips
BullBearCase	Analyst-style bull/bear thesis bullets
TradingViewWidget	Embedded chart (visual only; not wired to engine)
IndexBar	NIFTY 50, SENSEX, VIX live strip
SignalBadge	BUY / HOLD / SELL badge with colour
ScoreHistoryChart	Composite score trend over time
NewsCard	Sentiment-tagged news card
BacktestPanel	Walk-forward results per horizon on Picks page
LivePerformanceTracker	Per-pick P&L — entry, return%, alpha vs Nifty
PaperTradeModal	Trade form (qty, horizon, pre-filled SL/target)

Daily Picks Trust Layer

The Picks page has a collapsible **"Show Real Accuracy"** panel with three layers:

- **Backtest results** — real walk-forward hit rate, avg return, Sharpe, alpha vs Nifty per horizon
- **Confidence calibration table** — empirical hit rate per score band (60–65, 65–70, 70–75, 75–80, 80–85, 85–91) so users can see if higher confidence = higher win rate
- **Live P&L tracker** — every past daily pick with entry price, current return%, and alpha vs benchmark

Pick Card UI

- Rank badge (#1–#5)
- Score band label (STRONG BUY / BUY / HOLD) with colour
- Sector tag
- Top 3 signals inline (▲ BULLISH, ▼ BEARISH, → NEUTRAL) without needing to expand
- Compact market regime bar

21. Infrastructure & Deployment

Hosting

Layer	Provider	Plan
Backend API	Railway	Hobby (\$5/month, always-on, no cold starts)
Frontend	Vercel	Hobby
Database	PostgreSQL (Railway) or SQLite (local)	—
Auth	Supabase	Free tier

Migrated from Render to Railway (Session 5) — Railway Hobby plan eliminates the free-tier cold-start problem (30-second spin-up delays) and provides a persistent, always-on server.

Environment Variables

Variable	Description	Default
ENVIRONMENT	production / development	development
DATABASE_URL	PostgreSQL connection string	SQLite fallback
USE_POSTGRES	1 = Postgres, 0 = SQLite	0
PICKS_SECRET	Secret header for /api/picks/generate	Required in prod
PICKS_UNIVERSE_LIMIT	Cap stock count for picks run	25
PICKS_CANDIDATES	Top N from Phase-0 momentum screen for deep prediction	751 (set in Railway)
SCREEN_BATCH_SIZE	NSE bulk download batch size (memory safety)	300
MIN_MCAP_CR	Min market cap in ■ Cr for NSE universe	100
SCREENER_EMAIL	screener.in login (Indian fundamentals)	Required
SCREENER_PASSWORD	screener.in password	Required
FRONTEND_URL	Vercel frontend URL for CORS	Must be set in prod

Backend Startup Sequence

- Uvicorn starts FastAPI app
- CORS middleware enabled (frontend HTTPS origin)
- **Postgres schema initialised** (`init_db()` — creates all tables if not exist)
- **screener.in login** — authenticated session established on boot (not lazily)
- Prediction router with daemon-thread async pattern initialised

- Universe refresh (async, 30s delay)
- Keepalive loop (14-minute self-ping interval)
- Outcome resolver (6-hour interval)
- Warmup loop — pre-computes RELIANCE:IN:medium and AAPL:US:medium (150s delay, 90s gap)

22. Automation Workflows

Directory: [.github/workflows/](#)

daily_picks.yml — Daily Picks Generation

Schedule: 0 3 * * 1-5 (3:00 AM UTC = 8:30 AM IST, Mon-Fri)
Step 1: Poll /health up to 10× (30s gap) until Render responds 200
— prevents silent failure when server is cold-starting
Step 2: POST /api/picks/generate with x-secret header
Purpose: Ensures picks are generated even after overnight sleep
Result: Picks ready by 9:00-9:10 AM IST

weekly_validation.yml — Model Accuracy Validation

Schedule: 0 2 * * 0 (2:00 AM UTC = 7:30 AM IST, Sunday)
Steps: 1. Ping /health to wake Render
2. POST /api/validation/run?horizon=medium → wait 15 min
3. POST /api/validation/run?horizon=long → wait 25 min
4. GET /api/validation/results → print summary
Output: BUY hit rate %, alpha %, Sharpe per horizon

keep_alive.yml — Render Free Tier Keep-Alive

Schedule: */10 * * * * (every 10 minutes, 24/7)
Action: GET /health
Purpose: Prevent Render free tier from spinning down

23. Persistence & Data Durability

Render's free tier uses ephemeral disk — files written locally are wiped on every restart/redeploy. All user-facing and learning data is stored in PostgreSQL to survive this.

What Lives in Postgres

Data	Table	Survives Restart
Price alerts	price_alerts	■ Yes
Watchlist	watchlist	■ Yes
Paper trades & portfolio	paper_trades , paper_portfolio	■ Yes
Daily picks cache	daily_picks_cache	■ Yes
Validation results	val_runs , val_signals	■ Yes
Alpha engine predictions	predictions	■ Yes

Data	Table	Survives Restart
Outcome resolution	<code>outcomes</code>	■ Yes
IC history	<code>factor_ic_history</code>	■ Yes
Regime log	<code>regime_log</code>	■ Yes
Score snapshots	<code>score_snapshots</code>	■ Yes

What Is Transient (acceptable)

Data	Storage	Why Acceptable
Trained ML models (<code>meta_model_*.pkl</code> , <code>regime_kmeans.pkl</code>)	Local file	Auto-retrains from Postgres on next run — one cycle of degraded weights, no user data lost
API response caches (quotes, heatmap, movers)	In-memory (TTL)	Market data; freshly fetched anyway

screener.in Session

- Login fires at Render boot (not lazily on first request)
- Session refreshed every 6 hours
- `SCREENER_EMAIL` + `SCREENER_PASSWORD` must be set as Render environment variables
- Login logs: `[startup] screener.in login succeeded/failed` — check Render logs after deploy

24. Factor Weights by Horizon

Short-Term (1–5 days)

Factor	Weight	Rationale
Technical	70%	Momentum and price action dominate short windows
Fundamental	15%	Slow-moving signal, limited short-term relevance
Sentiment	15%	News-driven price moves matter in 1–5 days
Quality	Not applied	Computation overhead; not predictive short-term

Medium-Term (2–4 weeks)

Factor	Weight	Rationale
Technical	40%	Trend still matters but less dominant

Factor	Weight	Rationale
Fundamental	45%	Growth and valuation start to drive returns
Sentiment	15%	News flow still relevant over multi-week horizon
Quality	Included in composite	Institutional signals begin to matter

Long-Term (3–6 months)

Factor	Weight	Rationale
Technical	15%	Mean reversion reduces technical edge
Fundamental	75%	Business quality and valuation dominate
Sentiment	10%	Structural, not ephemeral, news matters
Quality	Included, weighted heavily	Piotroski, ROIC, cashflow most predictive

25. Key Design Principles

- **No Look-Ahead Bias** — Backtester uses only data available at the prediction date. Forward returns computed strictly after prediction timestamp.
- **Full Explainability** — Every signal has reasoning bullets, factor breakdown, and confidence components. Nothing is a black box.
- **Honest Risk-Adjusted Returns** — Target prices are not inflated. Trade levels (entry, stop, target) are mutually consistent with the signal.
- **Sector-Aware Scoring** — Different valuation thresholds for banks (ROE > 15%, P/B < 1.5), NBFCs, and IT companies vs industrials.
- **Institutional-Grade Signals** — Piotroski F-Score, Sharpe Ratio, ROIC, IC engine, Ledoit-Wolf covariance optimisation — same tools used by quant funds.
- **Data Resilience** — Three-layer fallback chain: yfinance → screener.in → BSE API. If news is unavailable, weights redistribute gracefully.
- **Memory-Efficient** — Designed for Render's 512 MB free tier. Cache capped at 300 entries with LRU eviction. Concurrent predictions use daemon threads, not async tasks.
- **Self-Improving** — Outcome logger tracks every prediction. IC engine retrains weekly. Factor weights evolve as the model sees more real-world outcomes.
- **Real-Time Ready** — 15-minute prediction cache, async background computation, React Query polling for live data.

- **Investor Transparency** — Every number in the UI is traceable to a specific calculation in the codebase. This document is kept current with every code change.

- **Postgres-First Persistence** — All user data (watchlist, alerts, paper trades, picks, validation, alpha engine) is stored in PostgreSQL. Render's ephemeral disk is never trusted for user-facing state.

26. Changelog

Session 5 — 2026-06-20

User Identity & Persistence:

- **All features migrated to Supabase `user_id`** — watchlist, alerts, paper trading, and StockContextMenu all previously used a hardcoded `USER_ID = "default"` or a localStorage `session_id`. Every feature is now scoped to the authenticated Supabase user UUID (`useAuth().user.id`), making data persist correctly across all browsers and devices.
- **Paper trading backend rewritten** — all 5 API endpoints migrated from `session_id` (random localStorage UUID) to `user_id`. `BuyRequest`, `SellRequest`, `EditRequest` models updated; `_ensure_portfolio()` queries by `user_id`. Old trades with `user_id = NULL` are legacy-only.
- **Dashboard watchlist fixed** — was fetching `/api/watchlist/default`; now uses `userId` from `useAuth`.

Heatmap Expansion & Quality:

- **India: 18 → 25 sectors** — added Healthcare, Insurance, Chemicals, Cement, Metal & Mining, Defence, Realty, Telecom, Consumer Disc, Hotels & Travel, Food & Beverage, Media & Entmt, Textiles, Agro & Chemicals, Logistics, Paints, Infra, Capital Goods, Power, EV & New Energy
- **US: 13 → 29 sectors** — added Cybersecurity, Fintech, Biotech, Med Devices, Clean Energy, EV, Consumer Stap, E-commerce, Social Media, Streaming & Media, Gaming, Airlines, Cruise & Hotels, Restaurants, Retail, Telecom, Utilities, Realty, Materials, Crypto & Blockchain
- **MAX_STOCKS raised 10 → 15** — wider coverage per sector
- **Full symbol audit** — all 353 India heatmap symbols bulk-tested via Yahoo Finance. 37 bad symbols identified; confirmed replacements applied (ATGL, ASTERDM, BAYERCROP, CANFINHOME, DALBHARAT, GUJGASLTD, ICICIPRULI, KNRCON, LEMONTREE, MFSL, MTARTECH, NH, ORIENTCEM, TEJASNET, TIPSFILMS, VTL, VIJAYA, VINATIORGA, WAAREEENER, WELSPUNLIV, ZENSARTECH, ETERNAL). Symbols with no Yahoo Finance listing removed (SPICEJET, GREENKO, HEXAWARE, KEYSTONE, etc.)
- **ZOMATO → ETERNAL** — company rebranded to Eternal on NSE
- **NSE `SECTOR_TO_NSE_INDEX` expanded** — 13 new sector → NSE index mappings added for primary data sourcing via NSE APIs

UX Fixes:

- **Landing page auth confusion fixed** — page now detects login state via `useAuth`; CTA switches from "Sign In" to "Go to Dashboard" when logged in; bottom CTA shows user email
- **Loading status badge** added to Market Overview (dashboard) and Heatmap — three states: blue "Fetching..." (first load), yellow "Refreshing..." (background poll), green wifi "Updated HH:MM:SS" (live)
- **M%26M / URL-encoding fix** — NSE symbols with `&` (M&M, M&MFIN) were displaying as `M%26M` in the stock page header. Fixed with `decodeURIComponent` on the stock page params and `encodeURIComponent` when building `/stock/` URLs in heatmap, context menu, and picks pages
- **Paper Trade modal unblocked** — Buy button was disabled until `fetchPrediction` completed (10–30s). Prediction now runs in background for stop loss/target suggestions only; button is immediately available

Infrastructure:

- **Migrated backend from Render → Railway Hobby** (\$5/month, always-on, no cold starts)
- **PICKS_CANDIDATES=751** set in Railway environment — expands the momentum screen candidate pool

Session 4 — 2026-06-19

Forensic Audit & Critical Fixes (9 issues resolved):

- **BUY threshold lowered 70 → 60** — the 70-point threshold was structurally unreachable for most NSE stocks on neutral/bearish market days. New thresholds: BUY ≥ 60 , HOLD 45–59, SELL < 45 . Score bands updated to match exactly (no more "Good Watchlist Stock" label on stocks treated as HOLD).
- **Fundamental score per-category budgets** — replaced unbounded additive accumulation (theoretical max ~215) with six capped buckets: valuation ± 15 , profitability ± 15 , growth ± 15 , balance sheet ± 10 , governance ± 10 , banking ± 10 . Scores now discriminate meaningfully across the full 0–100 range.
- **Growth double-counting fixed** — revenue and earnings growth were previously counted 3–5× simultaneously (TTM + 3Y CAGR + 5Y CAGR + trend, all additive). Now each counted once via the longest available window; quarterly trend is a capped supplement.
- **Quality gate fixed** — OCF check used Python `or` which treated zero cash flow as falsy; now uses explicit `is None`. Rejection logic split into independent OR conditions (was AND — too strict).
- **Sentiment denominator corrected** — neutral articles no longer dilute bullish signal. Denominator is now `bullish + bearish` only.
- **Race condition on `_generating` flag** — concurrent POST `/picks/generate` requests could both pass the guard simultaneously. Fixed with `threading.Lock`.
- **Optimizer fallback `max_weight` enforcement** — the fallback could allocate 100% to one stock when alphas were skewed. Fixed with iterative clipping loop.
- **Long-horizon IC training** — long horizon was training on 20D returns (1 month) despite a 3-6 month stated horizon. Now uses `return_60d`. Outcome logger waits 90 calendar days before resolving long-horizon predictions.
- **Partial returns never logged** — outcome logger previously logged partial forward returns when the window hadn't elapsed, corrupting IC training data. Now returns `None` until the full window is complete.
- **Validation thresholds synced** — validation BUY threshold (was 65) now matches live system (60) for all horizons.
- **`picks_generated_today()` logic fixed** — a prior run that produced 0 BUY signals (before threshold fix) saved an empty payload with today's date, causing the startup catch-up to skip regeneration. Now requires at least one actual pick to count as "done today".
- **NSE Daily Picks expanded** — universe expanded from Nifty 100 (96 stocks) to all NSE-listed stocks screened by market cap $\geq$ ₹100 Cr (~500-600 stocks). Two-phase pipeline: Phase-0 bulk momentum screen → Phase-1 deep prediction on top 50 candidates only (memory-safe batching of 300).
- **Documentation fully updated** — README and STOCKSENSE_DOCUMENTATION.md updated to reflect all threshold, formula, and architecture changes.

Live test results (2026-06-19):

- `/health`: 🟢 ok
- `/api/predictions/TCS?market=IN&horizon=medium`: 🟢 Score 78, BUY, Strong Buy Candidate
- `/api/screener/top-movers?market=IN`: 🟢 10 gainers, 10 losers
- `/api/validation/results?horizon=medium`: 🟢 Hit rate 56.6%, avg return 3.75%
- `/api/picks/status`: 🟢 Generating (startup catch-up triggered correctly)

Session 3 — 2026-06-18

New Features:

- **Paper Trading module** — full simulated trading with open/close positions, stop-loss/target tracking, unrealised and realised P&L, Postgres persistence
- **Price Alerts system** — Postgres-backed price level alerts with live trigger detection
- **Daily Picks Trust Layer** — real backtest results, confidence calibration table per score band, live P&L tracker for past picks
- **Pick Card UI overhaul** — rank badges (#1–#5), sector tags, top 3 signals visible inline, compact regime bar

Fixes:

- Watchlist migrated from ephemeral JSON file to Postgres — no longer disappears on Render restart
- screener.in login now fires at startup (not lazily) — data available from first request
- screener.in login enhanced: tries both `username/email` field names, logs every step
- Daily picks cron now waits for Render to be healthy before triggering — prevents silent cold-start failures
- Paper trading open positions: stop loss/target hint row now always visible (shows "not set — click ↗ to add one" when unset, for consistent layout)
- Open positions now always show SL/target hint row consistently regardless of signal type

Performance Improvements:

- `get_fundamentals`: replaced `3×sleep(3)` blocking retry with `asyncio.wait_for(timeout=8s)`
- OHLCV data: added 5-minute in-process cache (was completely uncached)
- Quote enrichment: removed redundant second yfinance `fast_info` call
- Heatmap cache TTL: 3 min → 5 min
- Dashboard refetch interval: 60s → 120s, staleTime aligned
- Heatmap frontend refetch: 3 min → 5 min, staleTime aligned
- Stock detail quote refetch: 30s → 60s
- `refetchOnWindowFocus`: `false` added across all major queries

Session 2 — (prior)

- Walk-forward validation engine with Postgres storage
- Learning Alpha Engine (IC weights, regime clustering, meta-model, outcome logger)
- Screener.in authenticated scraping for Indian fundamental data
- Portfolio page with BUY/HOLD/SELL signals per holding
- History tab with horizon selector on portfolio page
- Backtest async fix (non-blocking event loop)
- Prediction cache size cap (300 entries, LRU eviction)
- Hammer & Morning Star candlestick pattern bug fixes

Session 1 — (initial)

- Core prediction engine (technical, fundamental, sentiment, quality, macro)
- Stock detail page with full factor breakdown
- Dashboard with top movers (US/IN/Crypto)
- Heatmap page (sector-wise colour-coded)
- Screener with filters
- Watchlist with live prices
- Daily Picks engine (9-phase Learning Alpha pipeline)

- GitHub Actions automation (daily picks, weekly validation, keep-alive)
 - Render + Vercel deployment
-

This document is a living record of StockSense. It is updated with every significant change to the product.